



Qualification	Institute	Year	Result
• PhD Doctorate	Observatoire de Paris, Université PSL, France	2026 - 2029	—
• M2 Masters Plasma Physics & Fusion	Sorbonne Université, Université Paris Saclay & IPP Paris, France	2025 - 2026	12.04/20 Transcript
• M1 Masters Paris Physics Masters	Sorbonne Université & Université Paris Cité, France	2024 - 2025	12.33/20 Transcript
• BS-MS* *(4 th -year BS Exit)	Indian Institute of Science Education and Research Bhopal, India	2020 - 2024	8.15/10 Transcript

PUBLICATIONS

Dependence of spicule properties on magnetic field – results from magnetohydrodynamic simulations

KARTAV KESRI, S. Dey, P. Chatterjee, R. Erdelyi
The Astrophysical Journal

(Published: Sept. 17th, 2024)
DOI: [10.3847/1538-4357/ad67d8](https://doi.org/10.3847/1538-4357/ad67d8)

THESES

Ph. D. Thesis

October 2026 - September 2029

• Thesis title: *Towards a better understanding of the production and propagation of Jovian radio emissions through the magnetosphere*

• Advisors: Dr. Corentin Louis, Dr. Baptiste Cecconi

Laboratoire LIRA, Observatoire de Paris, UNIVERSITÉ PSL, PARIS, FRANCE

Master's Thesis

March 2026 - July 2026

• Thesis title: *Interactions of the solar wind with the Jovian magnetosphere*

• Advisors: Dr. Corentin Louis, Dr. James Waters

Laboratoire LIRA, Observatoire de Paris, UNIVERSITÉ PSL, PARIS, FRANCE

KEY RESEARCH EXPERIENCES

Matter–Antimatter Asymmetry from Gravitational Thermodynamics

July 2026 - Present

Dr. David S. Pereira | Assistant Professor

INSTITUTE OF ASTROPHYSICS AND SPACE SCIENCESM LISBON, PORTUGAL

Dependence of the properties of simulated solar spicules on imposed magnetic field

May 2023 - July 2023, Extended

Dr. Piyali Chatterjee | Associate Professor

INDIAN INSTITUTE OF ASTROPHYSICS, BANGALORE, INDIA

[Offer Letter](#) ◊ [Certificate](#)
[Project Report](#)

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GRANTS AND SCHOLARSHIPS

Year	Type	Details
2025	Grant	European Union's FuseNet Financial Grant for 62 nd Culham Plasma Physics Summer School, Oxfordshire, UK. Grant Letter
2024-25	Scholarship	Sorbonne Université Paris Physics Masters Scholarship. Certificate
2023-24	Fellowship	Visiting Student Fellowship at IUCAA, Pune, India.

2023	Fellowship	Visiting Student Fellowship at IIA, Bangalore, India.
2022-23	Scholarship	Uttar Pradesh Annual Need-Based Scholarship for Graduate Students.
2020-22	Scholarship	Central Sector Scholarship Scheme for Graduate Students under NSP.

CONFERENCES/WORKSHOPS

Year	Theme	Details (Topic, Activity, Location)
July 2026	Conference	Poster at MAGNETOSPHERES OF THE OUTER PLANETS, to be held at <i>Hotel Mercure</i> , Toulouse, France . Poster Link
July 2025	Workshop	4 th MANiTOU GRAVITATIONAL WAVES, held at <i>Aix-Marseille Université</i> , Marseille, France
June 2025	Workshop	KINETIC PHYSICS OF ASTROPHYSICAL PLASMAS, held at <i>Sorbonne Université</i> , Paris, France .
June 2025	Conference	Poster at 17 th WORKSHOP ON SOLAR INFLUENCES ON EVERYTHING held at <i>Primorsko Del Sol</i> , Primorsko, Bulgaria . Poster Link
May 2025	Workshop/ Discussion	X-SHOOTING ULLYSES (XShootU), <i>Institute of Science and Technology</i> (ISTA), Vienna, Austria .
May 2024	Workshop & Conference	Poster at THEORETICAL AND GEOPHYSICAL FLUID DYNAMICS, <i>ICTS-TIFR</i> , Bangalore, India . Poster Link
Feb 2024	Conference	Poster at 42 nd ANNUAL ASTRONOMICAL SOCIETY OF INDIA CONFERENCE (ASI '24), held at <i>Indian Institute of Science</i> , Bangalore, India . Poster Link
Jan 2024	Workshop	CORONAL MASS EJECTIONS, as part of ASI 2024 held at <i>Indian Space Research Organization</i> (ISRO), India .
Sept 2023	Discussion	ADITYA-L1, <i>Indian Institute of Technology Kanpur</i> , Kanpur, India
July 2023	Workshop	COMPUTATIONAL MHD FLUID SIMULATIONS USING PLUTO CODE, <i>Indian Institute of Science</i> , India

SKILL SETS

Coding Languages : C, C++, Python

Pipelines (Med. exp.): PLUTO CODE, GNUPLOT

Softwares/Environ. : ParaView, Wolfram Mathematica, GitHub, Linux Commands (Ubuntu 20.04), Parallel Processing (basics of MPI4Py), $\text{\LaTeX}$

Proficiency : French (A1), English ([TOEFL Scores](#)), Hindi & Awadhi (Mother Tongue(s))

KEY COURSES TAKEN

At Sorbonne Université, Paris, France

Physics: Diffusive Astrophysical Plasmas, Relativistic Astrophysics, Space Plasmas, High Energy Density Astrophysics, Tools for Plasma and Fusion, Magnetohydrodynamics, Kinetic Theory of Plasma, Waves and Instabilities, Numerical methods, Instrumentation, Radiation Physics, Plasma Physics, Advanced Quantum Mechanics, Statistical Mechanics.

At IISER Bhopal, India

Physics: Introduction to Astronomy and Astrophysics, Cosmology I, Atomic and Molecular Physics, Nuclear and Particle Physics, Electrodynamics and STR, General Theory of Relativity, Condensed Matter Physics, Mathematical Methods I & II, Quantum Mechanics I & II, Statistical Mechanics, Numerical Methods and Programming, Classical Mechanics, Thermal Physics, General Properties of Matter, Electromagnetism, Waves and Optics, Quantum Physics, Mechanics.

!! [Click here to see the complete course history](#) !!

Last Updated: July 29th, 2026